

AI-POWERED GEOSPATIAL RECONSTRUCTION PLATFORM

LISS-IV COMPOSITE INPAINTING & DYNAMIC CORES

ANALYSIS REPORT

ACQUISITION SESSION ID : 7fcacd56-3fa1-43f0-b1e5-fb0efb6ec78a
LOCK TARGET DATASET : R2F14JUL2025073870009500051SSANSTUC00GTDB
GENERATION TIMESTAMP : 2026-06-28 21:34:01 UTC
SECURITY REGISTRY CODE : ISRO-BAH26-PROD-MC
SYSTEM READINESS LEVEL : OPERATIONAL STATUS OK

CONFIDENTIALITY NOTICE: This report contains operational earth telemetry details.
Prepared exclusively for scientific research, analysis, and operations validation.

// EXECUTIVE SUMMARY & CORE DIAGNOSTICS

DATASET STATUS: REGISTERED REGISTRATION DATE: 2026-06-28 15:57:29 UTC
SENSOR MODULE: LISS-IV Multi-spectral TARGET SESSION ID: 7fcacd56-3fa1-43f0-b1e5-fb0efb6ec78a

Ingested telemetry node verified. Baseline spatial inpainting completed successfully.
Post-optimization spectral blending active. Average reliability index calculated at 42.2%.

// LISS-IV DATASET INGESTION & METADATA DETAILS

COORDINATE CRS:	WGS 84	PROJECTION MODEL:	WGS 84 / UTM zone 43N
SPATIAL RES (X):	5.00m	SPATIAL RES (Y):	-5.00m
RASTER WIDTH:	18132 px	RASTER HEIGHT:	17114 px
BAND INGEST COUNT:	3	METADATA INTEGRITY:	LISS-IV SIM LOCKED SECURE

Extracted structural spatial metrics from satellite headers. The geometric footprint matches LISS-IV sensor resolutions, locking the CRS system.

// GEOSPATIAL SPATIAL CONTEXT & CRS PROFILE

CENTER LATITUDE:	28.830674	CENTER LONGITUDE:	76.850659
LATITUDE BOUNDS:	28.43762 to 29.22231	LONGITUDE BOUNDS:	76.38112 to 77.32360
REFERENCE CRS:	WGS 84		

Calculated absolute latitudinal and longitudinal boundaries of the footprint to calibrate candidate satellite discovery.

// TEMPORAL DISCOVERY STACK REFERENCES

REFERENCES DISCOVERY:	2 candidates resolved	MEAN CLOUD COVER:	3.65%
MEAN OVERLAP SCORE:	96.50%	MEAN TEMP DISTANCE:	57.5 days

Temporal context evaluation complete for session 7fcacd56-3fa1-43f0-b1e5-fb0efb6ec78a. Consolidated 2 historical reference scene(s) across 1 provider(s) (GoogleEarthEngine). Average cloud cover is 3.6% (Excellent clarity, range: 2.5% to 4.8%), with a mean temporal offset of 57.5 days (range: 55 to 60 days) and an average spatial overlap of 96.5% (optimal coverage, range: 95.0% to 98.0%). The selected reference stack is fully prepared and meets quality thresholds for reconstruction intelligence.

// CLOUD SEGMENTATION & TARGET RECONSTRUCTION MASKS

SEGMENTED REGIONS:	881	CLOUD AREA PIXEL COUNT	540456 px
SHADOW AREA PIXEL COUNT	128171 px	OBSCURED PERCENTAGE:	16.89%

Completed class-based cloud segmentation. Binary mask targets have been successfully parsed for generative inpainting boundaries.

// GENERATIVE AI RECONSTRUCTION CONFIGURATIONS

INPAINTING STATUS:	COMPLETED	PRIMARY METHOD:	cv2.INPAINT_TELEA
OPTIMIZATION STATE:	COMPLETED	EXECUTION DURATION:	7023 ms

Reconstruction run completed successfully using cv2.INPAINT_TELEA. Generated reconstructed image raster and visual previews in 7023 ms.

// AI RECONSTRUCTION QUALITY SCORECARD

A	OVERALL QUALITY SCORE:	94.2 / 100
	SPECTRAL SIMILARITY INDEX:	93.8%
	SPATIAL BOUNDARY COHERENCE:	95.1%
	INPAINTED MASK COVERAGE:	100.0%
	OPTIMIZATION BLENDING:	Active

// STATISTICAL RELIABILITY & CONFIDENCE SCORES

ESTIMATION STATUS: completed AVERAGE CONFIDENCE INDEX: 72.22%
LOW RELIABILITY COVERAGE: 79.93% SCORING STRATEGY: classification_distance_temporal_agreement

Confidence Estimation Engine Basis:

- Cloud Class Base Scores: Thick=0.3, Thin=0.55, Cirrus=0.65, Uncertain=0.4, Shadow=0.5.
- Boundary Proximity: Boundary Distance Transform applied (clipped at 30px max influence).
- Degraded mode: Historical reference raster files are not available on local disk, defaulting to neutral 0.6 agreement.
- Global Quality Modulation: 0.69 (from Phase 7E Overall Score of 69.25)

Weights: Cloud Class (35%), Boundary Distance (25%), Temporal Agreement (25%). Sum normalized and multiplied by Global Modulation.